

Kariminejad Neurodevelopmental Syndrome (KAREVS) — Comprehensive Disease Characterization

Disease: Kariminejad Neurodevelopmental Syndrome (KAREVS) **MONDO:** MONDO:0975795 · **OMIM:** #620937 · **Gene:** *RBSN* (rabenosyn-5 / ZFYVE20; OMIM *609511; HGNC:20759) **Category:** Mendelian, autosomal recessive **Evidence base:** Predominantly human clinical (2 families, 6 individuals) plus in-vitro/cell-biology and model-organism data on *RBSN*/rabenosyn-5 function.

Note on evidence strength. KAREVS is an ultra-rare, recently delineated Mendelian disorder described in only two families. Consequently, most clinical statements derive from small aggregated case-level reports (OMIM/primary papers), and many template fields (formal prevalence, survival statistics, QoL instruments, trials) have **no published data**; these are marked *Not available*. Mechanistic claims are supported by direct functional studies of *RBSN* in patient cells and model organisms.

Summary (Answer to the Research Question)

Kariminejad Neurodevelopmental Syndrome (KAREVS) is an autosomal recessive neurodevelopmental/neuromuscular disorder caused by **biallelic missense variants in the FYVE domain of *RBSN*** (rabenosyn-5), an endosomal Rab4/Rab5 effector. It is characterized by global developmental delay/mild-to-moderate intellectual disability, progressive muscle weakness (myopathy, sometimes with sensorimotor neuropathy or spastic paraplegia), distinctive facial dysmorphism, ophthalmoplegia/ocular anomalies, and variable, nonspecific brain-imaging abnormalities (leukoencephalopathy/delayed myelination). Mechanistically, the FYVE variants abolish binding to phosphatidylinositol-3-phosphate (PI3P), preventing rabenosyn-5 recruitment to early endosomes and thereby **delaying endosomal maturation and impairing endolysosomal degradation while sparing endosomal recycling** — a "separation-of-function" defect. Management is entirely supportive; recurrence risk is 25% and consanguinity is the principal risk factor.

1. Disease Information

Overview. KAREVS (OMIM #620937) is a rare autosomal recessive developmental disorder featuring global developmental delay (delayed walking by a few years, speech delay), impaired intellectual development, hypotonia and progressive muscle weakness, dysmorphic facial features, and variable nonspecific brain-imaging abnormalities. It was first delineated clinically by Kariminejad et al. (2015) in three Iranian siblings and molecularly resolved (gene = *RBSN*) by Paul et al. (2022) [PMID 26192890; 35652444].

P 35652444

"two distinct RBSN missense variants are responsible for a novel Mendelian disorder consisting of progressive muscle weakness, facial dysmorphisms, ophthalmoplegia and intellectual disability." — Paul et al., 2022

(P 35652444)

Key identifiers. - **OMIM:** #620937 (phenotype); *609511 (*RBSN* gene) - **MONDO:** MONDO:0975795 - **MedGen:** C5975371 - **Orphanet:** No dedicated ORPHAcode identified as of report date (*Not available*) - **ICD-10:** No specific code; maps to Q87.8 / F79 (general dysmorphic/ID categories) (*no dedicated code*) - **ICD-11:** No specific code; would fall under LD2F.1Y / 6A00 classes (*no dedicated code*) - **MeSH:** No dedicated MeSH heading (*Not available*)

Synonyms / alternative names. Kariminejad neurodevelopmental syndrome; **KAREVS;** "Intellectual disability, muscle weakness and characteristic face" (original descriptive title, 3p24.3–p25.3-linked recessive syndrome). Not to be confused with **Kariminejad-Nafissi syndrome** (a distinct skeletal dysplasia).

Data provenance. Disease-level aggregated resources (OMIM, MONDO, MedGen) built from **individual patient case reports** (2 families, n=6). No EHR/registry-scale data exist.

2. Etiology

Causal factor — genetic (Mendelian, monogenic). KAREVS is caused by **homozygous missense variants in *RBSN***. No environmental, infectious, or multifactorial etiology is involved.

Genetic risk factors. - Causal variants: **c.547G>A (p.Gly183Arg)** — Iranian family; **c.538C>G (p.Arg180Gly)** — Canadian Cree family (both NM_022340.3, FYVE domain) [PMID 35652444]. - **Consanguinity** is the dominant risk factor: both families are consanguineous and homozygous by descent. - Being a heterozygous carrier of a pathogenic *RBSN* allele confers risk of affected offspring only when both parents carry it (AR). - Modifier genes: none identified; phenotypic variability (below) is currently unexplained (*candidate modifiers not studied*).

Environmental / lifestyle / occupational risk factors. *Not applicable* — no non-genetic contributors are known or expected for a monogenic recessive disorder.

Protective factors. None described. In principle, absence of a second pathogenic allele (heterozygosity) is fully protective (carriers are unaffected) [PMID 26192890].

Gene–environment interactions. *None described.*

3. Phenotypes

All phenotypes derive from n=6 patients; "frequencies" are qualitative/small-count estimates from Kariminejad 2015 (Iranian family, F1) and Paul 2022 / OMIM clinical synopsis (Cree family, F2).

Phenotype	Type	Onset	Severity/ Course	Frequency	HPO
Intellectual disability (mild–moderate; IQ 42, 53)	Cognitive/ behavioral	Childhood	Stable–mild, lifelong	All (6/6)	HP:0001249
Global developmental delay (motor, speech)	Sign	Infancy/ early childhood	Walking 18– 25 mo	All	HP:0001263 / HP:0000750 / HP:0001270
Progressive muscle weakness / myopathy	Sign	Childhood	Progressive	All	HP:0003198 / HP:0001324
Hypotonia	Sign	Early	Variable	Frequent	HP:0001252
Distinctive facies (highly arched eyebrows, downslanting palpebral fissures, prominent nasal bridge/nose, columella below alae nasi, narrow mouth, narrow/high palate, maxillary hypoplasia, small chin)	Physical	Congenital	Stable	All	HP:0002553; HP:0000494; HP:0000426; HP:0000160; HP:0000189; HP:0000347
Ophthalmoplegia / impaired eye abduction; ptosis; amblyopia; hypermetropia	Sign	Childhood	Variable	Subset (≥ 1 each)	HP:0000602; HP:0000508; HP:0000646; HP:0000540
Dental caries	Sign	Childhood	—	Reported	HP:0000670
Short stature	Growth	Childhood	—	Reported (OMIM)	HP:0004322
Leukoencephalopathy / delayed myelination / white-matter hyperintensities	Lab/ imaging	Childhood	Nonspecific	2/3 F1	HP:0002352; HP:0012448

Phenotype	Type	Onset	Severity/ Course	Frequency	HPO
Hippocampal atrophy; enlarged ventricles	Imaging	Childhood	—	Single patients	HP:0410170; HP:0002119
Spastic paraplegia	Sign	Childhood	Progressive	2/3 F2	HP:0001258
Sensorimotor peripheral neuropathy	Sign	Childhood	—	F2	HP:0007141
Elevated serum CK; elevated lactate; hyperlipidemia; microcephaly	Lab/ physical	Childhood	—	Single patients (F2)	HP:0003236; HP:0002151; HP:0003077; HP:0000252
Intention tremor	Sign	Childhood	—	1 patient	HP:0002080

"...mild intellectual disability, progressive muscle weakness, and characteristic facies... highly arched eyebrows, down-slanting palpebral fissures, prominent nasal bridge, prominent nose, columella extending below alae nasi, narrow mouth, narrow palate, and dental caries, and in one of them an inability to abduct the left eye." — Kariminejad et al., 2015

P 26192890

Quality-of-life impact. No formal QoL instruments (EQ-5D/SF-36/PROMIS) have been applied (*Not available*). Expected functional impact: lifelong intellectual disability limiting independent living, plus progressive weakness/spasticity affecting mobility and self-care.

4. Genetic / Molecular Information

- **Causal gene:** *RBSN* (Rabenosyn, RAB effector; aliases **ZFYVE20**, RBSN5). HGNC:20759; NCBI Gene 64145; Ensembl ENSG00000131381; UniProt **Q9H1K0**; OMIM *609511; cytoband 3p25.3.
- **Pathogenic variants (KAREVS):**
- **c.547G>A, p.Gly183Arg** (homozygous; Iranian F1)

- **c.538C>G, p.Arg180Gly** (homozygous; Cree F2)
- Both are **missense**, in the **FYVE domain, germline**, homozygous; classified **pathogenic/likely pathogenic** (ACMG: segregation, functional evidence, extreme rarity). p.Arg180Gly is present only **once in gnomAD** (allele frequency $<1 \times 10^{-5}$); p.Gly183Arg is absent/ultra-rare [PMID 35652444].
- **Functional consequence:** selective (separation-of-function) loss of function — loss of PI3P binding → failure of early-endosome localization → impaired endosomal maturation/lysosomal degradation with **preserved recycling** [PMID 35652444].
- **Allelic series (distinct RBSN disorders):**
- **p.Gly425Arg** (ZFYVE20 c.1273G>A): severe multi-organ disorder — intractable seizures, developmental delay, microcephaly, dysostosis, osteopenia, dysmorphism, macrocytosis/megaloblastoid erythropoiesis, transient cobalamin deficiency, hypertriglyceridemia, partial cathepsin D deficiency, 50% reduced transferrin uptake (recycling defect) [PMID 25233840].
- **Biallelic loss-of-function: MFANDO** — Myelofibrosis, Congenital, with Anemia, Neutropenia, Developmental delay, and Ocular abnormalities (OMIM #620939). Caused by homozygous **c.289G>C, p.(Gly97Arg)** (NM_022340.3), which actually disrupts splicing of exon 5 → **absence of intact RBSN** (true loss of function) with larger, clustered EEA1-positive endosomes in patient fibroblasts. Three consanguineous sibs: congenital progressive myelofibrosis (reticulin fibrosis by 4 weeks), anemia, severe congenital neutropenia (filgrastim-refractory), thrombocytopenia, developmental delay, ocular anomalies, dysmorphism; proband died at 20 months with 46,XY complete sex reversal and sensorineural hearing loss; **hematopoietic stem-cell/bone-marrow transplant restored blood counts** in surviving sibs (Magoulas et al., 2018,

P 29784638

This is the key allelic contrast: **KAREVS lacks hematologic disease** and is caused by hypomorphic FYVE missense alleles, whereas MFANDO is a true-LoF, hematologic/multisystem disorder.

"distinct germline mutations in RBSN cause non-overlapping phenotypes with specific and discrete endolysosomal cellular defects." — Paul et al., 2022

([P 35652444](#))

- **Somatic vs germline:** germline only; no somatic/cancer association in patients (though complete loss is a tumor-suppressor phenotype in *Drosophila*).
- **Modifier genes / epigenetics:** none identified (*Not available*).
- **Chromosomal abnormalities:** none — disease is a point-mutation (missense) disorder; CMA/karyotype normal.

Ontology. Gene product: FYVE-domain Rab effector. GO-MF: phosphatidylinositol-3-phosphate binding **GO:0032266**; Rab GTPase binding **GO:0017137**. CHEBI: phosphatidylinositol 3-phosphate **CHEBI:26034**.

5. Environmental Information

Not applicable. KAREVS is a monogenic recessive disorder with **no** established environmental factors, lifestyle contributors, or infectious agents. No toxin/radiation/occupational exposures are implicated (searches of CTD/PubMed yield no environmental modifiers).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Homozygous FYVE-domain missense variant** (p.Gly183Arg or p.Arg180Gly) in *RBSN* **results in** an amino-acid substitution at a conserved residue of the phosphoinositide-binding FYVE finger. (*Demonstrated — genetics/segregation.*)
2. The altered FYVE domain **abolishes binding to phosphatidylinositol-3-phosphate (PI3P)**. (*Demonstrated — biochemistry.*)

3. Loss of PI3P binding **prevents translocation/recruitment of rabenosyn-5 to early endosomal membranes** (mutant protein fails to co-localize with EEA1). (*Demonstrated — cell imaging in patient fibroblasts.*)
4. Endosome-unassociated rabenosyn-5 **cannot properly scaffold the Rab5–VPS45–SNARE fusion/maturation machinery, leading to delayed early-to-late endosome maturation** and impaired delivery of cargo to lysosomes. (*Demonstrated — cargo tagged for lysosomal degradation accumulates.*)
5. Critically, the **endosomal recycling arm is spared** (separation-of-function) — the pathway branches here, distinguishing KAREVS from the G425R recycling-defect disorder. (*Demonstrated.*)
6. Impaired endolysosomal degradation **results in** accumulation of undegraded cargo and disturbed membrane/receptor homeostasis in **post-mitotic, trafficking-dependent cells (neurons, myofibers)**. (*Inferred from cellular data → tissue phenotype.*)
7. This **leads to** neuronal/white-matter dysfunction (→ intellectual disability, leukoencephalopathy, delayed myelination) and myofiber dysfunction (→ progressive myopathy; ± neuropathy/spastic paraplegia), plus developmental effects on craniofacial patterning (→ dysmorphism). (*Inferred.*)

Detail by category

- **Molecular pathways:** Rab5/Rab4 GTPase signaling on early endosomes; PI3-kinase→PI3P phosphoinositide signaling; SNARE-mediated membrane fusion via VPS45 (Sec1/Munc18). Not a classic Wnt/MAPK/mTOR disorder, though endolysosomal trafficking intersects growth-factor-receptor signaling.
- **Cellular processes:** early endosome fusion/maturation; receptor-mediated endocytosis; endosome-to-lysosome (degradative) transport; endosomal recycling (spared). Rabenosyn-5 *"is recruited in a phosphatidylinositol-3-kinase-dependent fashion to early endosomes... complexed to the Sec1-like protein hVPS45"* [PMID 11062261].
- **Protein dysfunction:** selective/partial loss of function — the protein is expressed but mislocalized; a hypomorphic/separation-of-function mechanism (not aggregation or dominant-negative). Complete loss is embryonic-lethal, so disease requires residual function.
- **Metabolic changes:** secondary and variable — elevated lactate and CK in some patients (suggesting mitochondrial/muscle stress); hyperlipidemia in one; the allelic G425R disorder shows cobalamin deficiency/hypertriglyceridemia (trafficking of nutrient receptors).

- **Immune involvement:** none in KAREVS (no hematologic/immune phenotype; contrast RBSN congenital myelofibrosis).
- **Tissue damage mechanisms:** inferred chronic cellular trafficking stress in neurons and myofibers; no fibrosis/ischemia mechanism.
- **Biochemical abnormalities:** loss of PI3P–FYVE interaction; partial cathepsin D processing defect described for the allelic G425R variant.
- **Epigenetic changes / omics profiling:** *Not available* (no transcriptomic/proteomic/metabolomic/single-cell datasets published for KAREVS).

Cell types (CL): neuron **CL:0000540**; skeletal muscle fiber **CL:0000188**; fibroblast (model) **CL:0000057**; oligodendrocyte (myelination, inferred) **CL:0000128**. **Biological process (GO):** endosome to lysosome transport **GO:0008333**; early endosome to late endosome transport **GO:0045022**; endosomal transport **GO:0016197**; regulation of endosome size/maturation; vesicle fusion **GO:0006906**.

7. Anatomical Structures Affected

- **Organ/system level (primary):** **central nervous system** (brain — cerebral white matter, hippocampus, ventricles) and **skeletal muscle** (neuromuscular system). Secondary/associated: **peripheral nerves** (sensorimotor neuropathy), **eye/extraocular muscles** (ophthalmoplegia, ptosis, refractive error), **craniofacial skeleton** (dysmorphism, maxillary hypoplasia), **skeletal growth** (short stature).
- **Body systems:** nervous (central + peripheral), musculoskeletal, ophthalmologic. **Not** primarily cardiovascular, respiratory, renal, or hematologic (distinguishing from allelic RBSN disorders).
- **Tissue/cell level:** nervous tissue (neurons, myelinating glia), striated muscle tissue (myofibers). Ubiquitously expressed gene, but post-mitotic trafficking-dependent tissues are most vulnerable.
- **Subcellular level:** **early endosome** (primary site of RBSN action) — GO cellular component early endosome **GO:0005769**, early endosome membrane **GO:0031901**; downstream **lysosome** **GO:0005764**; cytoplasmic vesicle. UniProt localizes rabenosyn-5 to the cytoplasmic face of early endosomes.
- **Localization / UBERON:** brain **UBERON:0000955**; cerebral white matter **UBERON:0002316**; hippocampus **UBERON:0002421**; lateral ventricle **UBERON:0002285**; skeletal muscle

UBERON:0001134; peripheral nerve UBERON:0001021; extraocular muscle UBERON:0002417.

- **Lateralization:** generally **bilateral/symmetric** (developmental/systemic); the original ocular finding (impaired abduction) was reported as unilateral (left eye) in one patient.

8. Temporal Development

- **Onset: congenital / early-childhood**, insidious. Motor and speech delay evident in infancy; independent walking delayed to 18–25 months [PMID 26192890].
- **Progression: chronic and slowly progressive**, principally the neuromuscular component ("*progressive muscle weakness*"); intellectual disability is relatively static. Spastic paraplegia (Cree family) is progressive.
- **Disease course:** progressive/stable hybrid — cognitive impairment stable, motor/neuromuscular features progressive. No episodic/relapsing pattern.
- **Duration:** lifelong. Survival into adulthood documented (Paul 2022 followed the original Iranian family long-term).
- **Remission:** none (no spontaneous or treatment-induced remission).
- **Critical periods:** early childhood (developmental/rehabilitation window); no proven pharmacologic intervention window.

9. Inheritance and Population

- **Epidemiology: ultra-rare.** Only **6 individuals from 2 families** reported worldwide (3 Iranian sibs; 3 Canadian Cree across 2 generations). Formal **prevalence/incidence not established** (consistent with $<1/1,000,000$).
- **Inheritance: autosomal recessive** (homozygous *RBSN* variants; unaffected consanguineous carrier parents) [PMID 26192890;

P 35652444

- **Penetrance:** appears **complete** in homozygotes (all reported homozygotes affected); carriers unaffected.

- **Expressivity: variable** — the Cree family expanded the phenotype (spastic paraplegia, neuropathy, ↑ CK/↑ lactate, microcephaly, hyperlipidemia, enlarged ventricles) beyond the Iranian family.
- **Genetic anticipation:** none (not a repeat-expansion disorder).
- **Germline mosaicism:** none reported.
- **Founder effects:** the two alleles are **family-private** (G183R Iranian; R180G Cree) — plausibly founder alleles within each consanguineous kindred rather than a broad population founder variant.
- **Consanguinity: central** — both families consanguineous; homozygosity mapping was the key discovery tool.
- **Carrier frequency:** not defined; RBSN pathogenic alleles are extremely rare in gnomAD (R180G seen once).
- **Population demographics:** reported in **Iranian** and **Canadian Indigenous (Cree)** populations. Sex ratio ~1:1 (both sexes affected). Age distribution: pediatric-through-adult.

10. Diagnostics

- **Definitive test: molecular genetic testing.** In consanguineous families, **homozygosity mapping + whole-exome (WES) or whole-genome (WGS) sequencing** identifies biallelic *RBSN* variants (confirmed by Sanger segregation) [PMID 26192890;

[P 35652444](#)

Single-gene/*RBSN*-inclusive neurodevelopmental or myopathy gene panels are appropriate when the phenotype is recognized.

"Using exome sequencing, we identified recessively acting germline alleles p.Arg180Gly and p.Gly183Arg." — Paul et al., 2022

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[P 35652444](#)

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- **CMA / karyotype / FISH:** expected **normal** (no CNV/structural/aneuploidy cause) — useful mainly to exclude mimics.

- **Mitochondrial DNA / repeat-expansion testing:** not indicated for causation, but mtDNA studies may be considered to exclude mitochondrial disease given elevated lactate/CK in some patients.
- **Supportive clinical tests:**
- **Electrophysiology (EMG/NCS):** myopathic pattern; sensorimotor neuropathy in some [PMID 26192890].
- **Muscle biopsy:** nonspecific.
- **Neuroimaging (MRI):** leukoencephalopathy, delayed myelination, frontal/parietal white-matter hyperintensities, hippocampal atrophy, enlarged ventricles (variable).
- **Labs:** serum CK and lactate may be elevated in a subset; no specific biomarker.
- **Omics-based diagnostics:** none validated (*Not available*); functional PI3P-binding/endosomal-localization assays are research tools that can support variant interpretation.
- **Clinical criteria:** no formal consensus criteria; diagnosis = compatible phenotype + biallelic pathogenic *RBSN* variants.
- **Differential diagnosis:** other AR syndromic intellectual disability with myopathy and dysmorphism; congenital/metabolic myopathies; mitochondrial encephalomyopathies (↑ lactate/CK); hereditary spastic paraplegias with cognitive involvement; and the **distinct RBSN-opathies** — the G425R seizure/microcephaly phenotype (P 25233840) and **MFANDO** congenital myelofibrosis with cytopenias (OMIM #620939, P 29784638).
- Distinguishing feature of KAREVS: characteristic facies + progressive myopathy **without** hematologic abnormality, with FYVE-domain *RBSN* genotype.
- **Screening:** no newborn-screening assay. **Carrier and cascade testing** feasible once the family variant is known; **prenatal / preimplantation genetic diagnosis** feasible for at-risk pregnancies.

NCIT: Whole Exome Sequencing; Whole Genome Sequencing; Genetic Testing; Magnetic Resonance Imaging; Electromyography.

11. Outcome / Prognosis

- **Survival / mortality:** No disease-specific mortality data. Reported patients **survived into adulthood**; life-limiting complications not documented in the small cohort (*formal survival statistics Not available*).
 - **Morbidity / disability:** Principal morbidity is **lifelong intellectual disability** plus a **progressive neuromuscular disability** (weakness ± spasticity/neuropathy) affecting mobility, and ocular/visual impairment. ICF-level functional limitations in mobility, communication, and self-care are expected.
 - **Quality-of-life measures:** none applied (*Not available*).
 - **Complications:** progressive weakness/contractures, spasticity, visual impairment; potential feeding/dental issues (narrow palate, caries). No organ failure described.
 - **Recovery potential:** none (non-remitting); rehabilitation can improve function but not reverse the underlying deficit.
 - **Prognostic factors:** genotype (allelic series predicts severity/organ involvement); presence of spastic paraplegia/neuropathy may portend greater motor disability. No validated prognostic biomarkers.
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12. Treatment

No disease-modifying or curative therapy exists. Management is **supportive, multidisciplinary, and symptom-directed** (extrapolated from standard care for AR syndromic ID/myopathy; no KAREVS-specific trials).

- **Pharmacotherapy:** no targeted drug. Symptomatic agents as needed (e.g., antispasticity agents such as **baclofen** for spastic paraplegia; standard management of any seizures). **Pharmacogenomics:** none specific.
- **Advanced therapeutics (gene/cell/RNA/targeted/immunotherapy): none available;** entirely theoretical. Gene-replacement/editing is conceptually plausible (recessive LoF) but faces the barrier that complete loss is lethal and CNS delivery is unsolved — no programs exist (*Not available*).
- **Surgical/interventional:** as indicated for complications (e.g., strabismus/ptosis surgery, orthopedic management of contractures).

- **Supportive & rehabilitative (mainstay): physical therapy** (weakness, spasticity, contracture prevention), **occupational therapy**, **speech-language therapy**, special education/developmental support, **ophthalmologic correction** (refractive error, amblyopia, ptosis/strabismus), **dental care**, and **nutritional support** as needed.
- **Experimental treatments / trials:** none registered (*Not available*).
- **Treatment outcomes / adverse events:** no cohort data.
- **Treatment strategy:** individualized supportive care coordinated by clinical genetics, neurology, physiatry, ophthalmology, and rehabilitation services.

NCIT: Physical Therapy; Occupational Therapy; Speech Therapy; Supportive Care; Rehabilitation Therapy; Baclofen.

13. Prevention

- **Primary prevention: genetic counseling** for consanguineous couples and families with an affected child; **carrier testing** of at-risk relatives; **preconception/prenatal counseling**. Recurrence risk to siblings = 25%.
- **Secondary prevention (early detection): cascade carrier screening** in the extended family; **prenatal diagnosis** or **preimplantation genetic testing (PGT-M)** once the familial *RBSN* variant is known.
- **Tertiary prevention:** early developmental intervention, physiotherapy to prevent contractures, proactive ophthalmologic and dental care to limit complications.
- **Immunization / public-health / environmental / behavioral / prophylaxis measures:** *Not applicable* (no infectious/environmental component). Standard childhood immunizations as per general guidelines.
- **Counseling:** autosomal-recessive genetic counseling is the cornerstone of prevention (NSGC/ACMG frameworks).

NCIT: Genetic Counseling; Genetic Carrier Screening; Prenatal Diagnosis.

14. Other Species / Natural Disease

- **Taxonomy:** No naturally occurring KAREVS-equivalent disease is reported in any non-human species (OMIA: none). Human — *Homo sapiens* (NCBI:txid9606).
 - **Orthologous genes (NCBI Gene / model):**
 - Mouse *Rbsn* (MGI:1925537; UniProt Q80Y56)
 - *Drosophila Rbsn* / CG8506
 - *C. elegans rabs-5*
 - *S. cerevisiae* functional homolog **VAC1**
 - **Natural disease in animals / veterinary relevance / breeds (VBO):** none known (*Not applicable*).
 - **Comparative biology:** the rabenosyn-5–VPS45–Rab5 endosomal-fusion module is **deeply conserved from yeast to humans**, so cellular disease mechanisms translate across species even though the human neurodevelopmental phenotype is not naturally observed elsewhere. "*Rabenosyn-5 is a closer mammalian functional homologue of yeast Vac1p than EEA1.*" [PMID 11062261].
 - **Transmission / zoonotic potential:** *Not applicable* (genetic, non-transmissible).
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15. Model Organisms

- **Cellular / in vitro (most relevant to KAREVS): patient-derived fibroblasts** (p.Gly183Arg) demonstrate the disease mechanism — loss of punctate endosomal localization, failure to co-localize with EEA1, and accumulation of lysosome-targeted cargo [PMID 35652444]. iPSC-derived neurons/myotubes are logical (not yet published) models.
- **Mouse (*Rbsn*, MGI:1925537): whole-body knockout is embryonic-lethal** — recapitulates the essential trafficking role but **not** the viable human phenotype; a **knock-in of the human missense (hypomorphic) allele or conditional/tissue-specific model** would be required to model KAREVS. No published KAREVS mouse to date.
- ***Drosophila* (*Rbsn*/CG8506):** null mutants show defective early-endosome formation, loss of epithelial polarity, and neoplastic overgrowth (*Rbsn* as tumor suppressor), establishing the *Rbsn*–Vps45–Rab5–Avalanche fusion module in vivo [PMID 18685079].

- ***C. elegans* (rabs-5 / vps-45):** mutants show endocytic defects with accumulation of aberrantly small endosomes, confirming RABS-5/VPS-45 cooperation downstream of RAB-5 [PMID 17235359].
- **Yeast (*VAC1*):** the functional homolog coordinates Rab/PI3K signaling in VPS45-dependent endosomal docking/fusion [PMID 11062261].
- **Phenotype recapitulation:** model organisms faithfully reproduce the **cellular/endosomal** mechanism but **not** the specific human neurodevelopmental/neuromuscular disease, because null alleles are lethal and human disease arises from residual-function missense alleles. **Model limitation:** need for allele-specific (knock-in) and CNS/muscle-conditional models.
- **Applications:** these models illuminate endosomal fusion/maturation, PI3P–FYVE biology, and Rab5-effector function; allele-specific models would enable therapy testing.
- **Resources:** MGI (mouse), FlyBase (*Drosophila*), WormBase (*C. elegans*), SGD (yeast), IMPC/IMSR.

Supported vs. Refuted Hypotheses

Supported: - KAREVS is caused by biallelic FYVE-domain *RBSN* missense variants (AR) — *strongly supported* (segregation in 2 families + functional data) [PMID 26192890;

P 35652444

- Mechanism is loss of PI3P binding → endosomal mislocalization → delayed endosomal maturation with spared recycling (separation-of-function) — *supported* (patient-cell biochemistry/imaging) [PMID 35652444]. - Distinct *RBSN* variants cause distinct disorders via distinct endolysosomal defects — *supported* [PMID 35652444;

P 25233840

Refuted / excluded: - Environmental, infectious, or chromosomal causation — *excluded* (purely monogenic). - Hematologic involvement as part of KAREVS — *excluded* in reported families (distinguishes from *RBSN* congenital myelofibrosis). - Complete-null mechanism — *refuted* (null is lethal; disease requires hypomorphic alleles).

Limitations and Future Directions

- **Very small n (6 patients, 2 families):** frequencies, penetrance, expressivity, prognosis, and QoL are provisional. No registry, natural-history study, or trials exist.
 - **No omics data** (transcriptomics/proteomics/metabolomics) for patient tissues.
 - **Phenotypic variability unexplained** (candidate modifier genes and genotype–phenotype rules need larger cohorts, e.g., via GeneMatcher).
 - **Priorities:** identify additional families; generate allele-specific mouse/iPSC-neuron/myotube models; define natural history; explore whether modulating endolysosomal maturation or the PI3P–FYVE interaction is therapeutically tractable.
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Key References (PMIDs)

- **26192890** — Kariminejad et al., 2015. Original clinical delineation; 3p24.3–p25.3 mapping (human clinical).
 - **35652444** — Paul et al., 2022. *RBSN* gene identification; FYVE variants; separation-of-function mechanism (human clinical + in vitro).
 - **25233840** — Stöckler et al., 2014. Distinct *RBSN*/*ZFYVE20* p.Gly425Arg disorder (human clinical + in vitro).
 - **29784638** — Magoulas et al., 2018 (*Blood*). MFANDO — syndromic congenital myelofibrosis from *RBSN* loss-of-function p.Gly97Arg splice variant; BMT restores hematology (human clinical + in vitro).
 - **11062261** — Nielsen et al., 2000. Rabenosyn-5 as Rab5 effector, FYVE/PI3K recruitment, hVPS45 complex, Vac1p homolog (in vitro).
 - **18685079** — Morrison et al., 2008. *Drosophila* *Rbsn*–*Vps45* endosomal-entry module; tumor suppressor (model organism).
 - **17235359** — Gengyo-Ando et al., 2007. *C. elegans* VPS-45/RABS-5 in RAB-5-dependent endocytosis (model organism).
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